

Symbols for elements

Each element has its own place in the table based on its atomic number and is known by its name, chemical symbol, and its relative atomic mass. This is the average of the mass of all atoms of an element allowing for their proportion of isotopes (atoms of an element with a different number of neutrons than normal).

Atomic number — 11 22.990 — Relative atomic mass

Chemical symbol — **Na**

Sodium — Name

Modern periodic table

The table today contains 118 elements, more than 90 of which occur naturally in some form. The rest have been chemically made, often for just fractions of a second, in laboratories. The latest four (115–118) were named in 2016.

6		9		10		11		12		Aluminum		Silicon		Phosphorus		Sulfur		Chlorine		Argon	
26	55.845	27	58.933	28	58.693	29	63.546	30	65.39	31	69.723	32	72.64	33	74.922	34	78.96	35	79.904	36	83.80
Fe		Co		Ni		Cu		Zn		Ga		Ge		As		Se		Br		Kr	
Iron		Cobalt		Nickel		Copper		Zinc		Gallium		Germanium		Arsenic		Selenium		Bromine		Krypton	
44	101.07	45	102.91	46	106.42	47	107.87	48	112.41	49	114.82	50	118.71	51	121.76	52	127.60	53	126.90	54	131.29
Ru		Rh		Pd		Ag		Cd		In		Sn		Sb		Te		I		Xe	
Ruthenium		Rhodium		Palladium		Silver		Cadmium		Indium		Tin		Antimony		Tellurium		Iodine		Xenon	
76	190.23	77	192.22	78	195.08	79	196.97	80	200.59	81	204.38	82	207.2	83	208.96	84	(209)	85	(210)	86	(222)
Os		Ir		Pt		Au		Hg		Tl		Pb		Bi		Po		At		Rn	
Osmium		Iridium		Platinum		Gold		Mercury		Thallium		Lead		Bismuth		Polonium		Astatine		Radon	
108	(277)	109	(268)	110	(281)	111	(272)	112	285	113	284	114	289	115	288	116	293	117	294	118	294
Hs		Mt		Ds		Rg		Cn		Nh		Fl		Mc		Lv		Ts		Og	
Hassium		Meitnerium		Darmstadtium		Roentgenium		Copernicium		Nihonium		Flerovium		Moscovium		Livermorium		Tennessine		Oganesson	
61	(145)	62	(150.36)	63	151.96	64	157.25	65	158.93	66	162.50	67	164.93	68	167.26	69	168.93	70	173.04	71	174.97
Pm		Sm		Eu		Gd		Tb		Dy		Ho		Er		Tm		Yb		Lu	
Promethium		Samarium		Europium		Gadolinium		Terbium		Dysprosium		Holmium		Erbium		Thulium		Ytterbium		Lutetium	
93	(237)	94	(244)	95	(243)	96	(247)	97	(247)	98	(251)	99	(252)	100	(257)	101	(258)	102	(259)	103	(262)
Np		Pu		Am		Cm		Bk		Cf		Es		Fm		Md		No		Lr	
Neptunium		Plutonium		Americium		Curium		Berkelium		Californium		Einsteinium		Fermium		Mendelevium		Nobelium		Lawrencium	

1894

Scottish chemist William Ramsay discovered the element argon. He later discovered three others—neon, krypton, and xenon—and showed how they formed a new group of elements, the noble gases.

1898

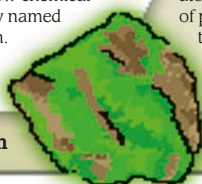
Polish-French scientist Marie Curie and French scientist Pierre Curie isolated two new chemical elements, which they named radium and polonium.

1913

English physicist Henry Moseley found that the nature of the X-rays an atom emits depends upon the number of protons inside it. This allowed the table to be organized by atomic number not relative atomic mass.

1940

American chemist Glenn Seaborg discovered plutonium and later nine elements that appear after uranium on the table. He proposed the addition of the actinide element series.



Pellet of radium